

ONLINE RETAIL BUSINESS INTELLIGENCE

Executive Decision Report & Capstone Presentation

Data-driven recommendations for revenue growth, customer retention and product optimization

Executive Summary

- Revenue analyzed: ₹7.17M equivalent dataset revenue
- 22.186K orders and ~4K customers are represented in the dashboard
- Average Order Value: 323.16
- Quantity and revenue show a very strong positive relationship ($r \approx 0.943$)
- Customer Churn Rate is a 90-day inactivity proxy: 33.26%
- Priority: protect high-performing products, reactivate inactive customers and focus commercial effort on high-value markets

Business Question & Objectives

Business questions:

- What drives revenue and order value?
- Which products and countries deserve commercial focus?
- Where is customer inactivity creating risk?
- Which actions can produce measurable financial impact?

Objectives:

- Convert transaction data into executive KPIs
- Identify revenue drivers and risk areas
- Recommend three practical, measurable interventions

Data & Analytical Methodology

Source: cleaned Online Retail transaction dataset

Workflow:

1. Data cleaning and preprocessing
2. Feature engineering: Year, Month, Month Name and Revenue
3. Descriptive statistics and outlier analysis
4. Correlation and hypothesis testing
5. Power BI KPI, trend, geographic and product analysis
6. Executive recommendations and scenario-based ROI modeling

Important: Churn is not directly recorded; 33.26% is an explicitly labeled 90-day inactivity proxy.

Executive KPI Snapshot

TOTAL REVENUE	7.17M
TOTAL ORDERS	22.186K
TOTAL CUSTOMERS	~4K
AVERAGE ORDER VALUE	323.16
90-DAY INACTIVITY / CHURN PROXY	33.26%

These KPIs form the executive monitoring layer of the Power BI dashboard.

Key Revenue Drivers

- Quantity vs Revenue: $r \approx 0.943$ — the strongest relationship identified
- Unit Price vs Revenue: $r \approx 0.018$ — very weak linear relationship
- Unit Price vs Quantity: $r \approx -0.022$ — weak inverse relationship
- Revenue fluctuates materially over time, with visible transaction-level spikes and negative-return periods

Decision implication:

Revenue growth should emphasize order volume, product demand and repeat purchasing rather than relying primarily on price increases.

Product Performance

The Top 10 Products by Revenue visualization identifies a concentrated group of high-value products.

Recommended executive action:

- Protect availability of top-revenue SKUs
- Prioritize these products in campaigns and cross-sell journeys
- Monitor stock-outs and return/cancellation behavior
- Build bundles around high-performing products to increase basket size

The dashboard should be used to refresh this ranking as new transaction data arrives.

Geographic Revenue Opportunity

The country-level map shows that revenue is geographically concentrated rather than evenly distributed.

Recommended action:

- Rank countries by revenue and order frequency
- Allocate marketing and inventory effort to high-value markets
- Use country slicers to compare performance over time
- Investigate low-revenue markets before increasing acquisition spend

The Power BI map provides the interactive layer for this decision.

Customer Retention Risk

33.26% of customers are classified as inactive under the dashboard's 90-day proxy definition.

Interpretation:

- This is a retention-risk signal, not contractual churn
- The metric can be used to create a reactivation target list
- Customers should be segmented by historical value before incentives are offered

Priority segments:

1. High-value inactive customers
2. Previously frequent customers showing recent inactivity
3. Low-value inactive customers requiring low-cost automated engagement

Three Strategic Recommendations

1. REACTIVATE HIGH-VALUE INACTIVE CUSTOMERS

Launch targeted email/offer campaigns for customers inactive >90 days.

2. PROTECT & GROW TOP PRODUCTS

Prioritize availability, promotion and cross-sell around the highest-revenue SKUs.

3. FOCUS COMMERCIAL EFFORT BY MARKET

Use country-level revenue and order data to concentrate marketing and inventory decisions on high-value markets.

ROI Scenario & Implementation Timeline

Illustrative scenario — not a financial forecast:

Baseline revenue: 7.17M

Assumed revenue uplift from combined initiatives: 5%

Illustrative incremental revenue: 358.5K

Illustrative implementation cost: 100K

Illustrative net benefit: 258.5K

Illustrative ROI: 258.5%

Timeline:

0–30 days: segment customers, validate top SKUs and market ranking

31–60 days: launch reactivation and product campaigns

61–90 days: optimize by response, revenue and margin outcomes

Executive Decision & Next Steps

Decision requests:

- Approve a 90-day customer reactivation pilot
- Protect inventory and promotion for top-revenue products
- Establish country-level performance reviews using the Power BI dashboard

Success measures:

- Revenue uplift
- Repeat-purchase rate
- Reactivation rate
- Average Order Value
- Product availability / return rate

Recommended cadence: weekly operational monitoring + monthly executive review.

Power BI Dashboard / Project Repository:

<https://github.com/sudeshna52/OnlineRetail>

Portfolio includes:

- Data cleaning and preprocessing
- Exploratory data analysis
- Power BI interactive dashboard
- Executive KPI and drill-down analysis
- PBIX and PDF dashboard deliverables

This report synthesizes the analytical work into an executive decision layer.